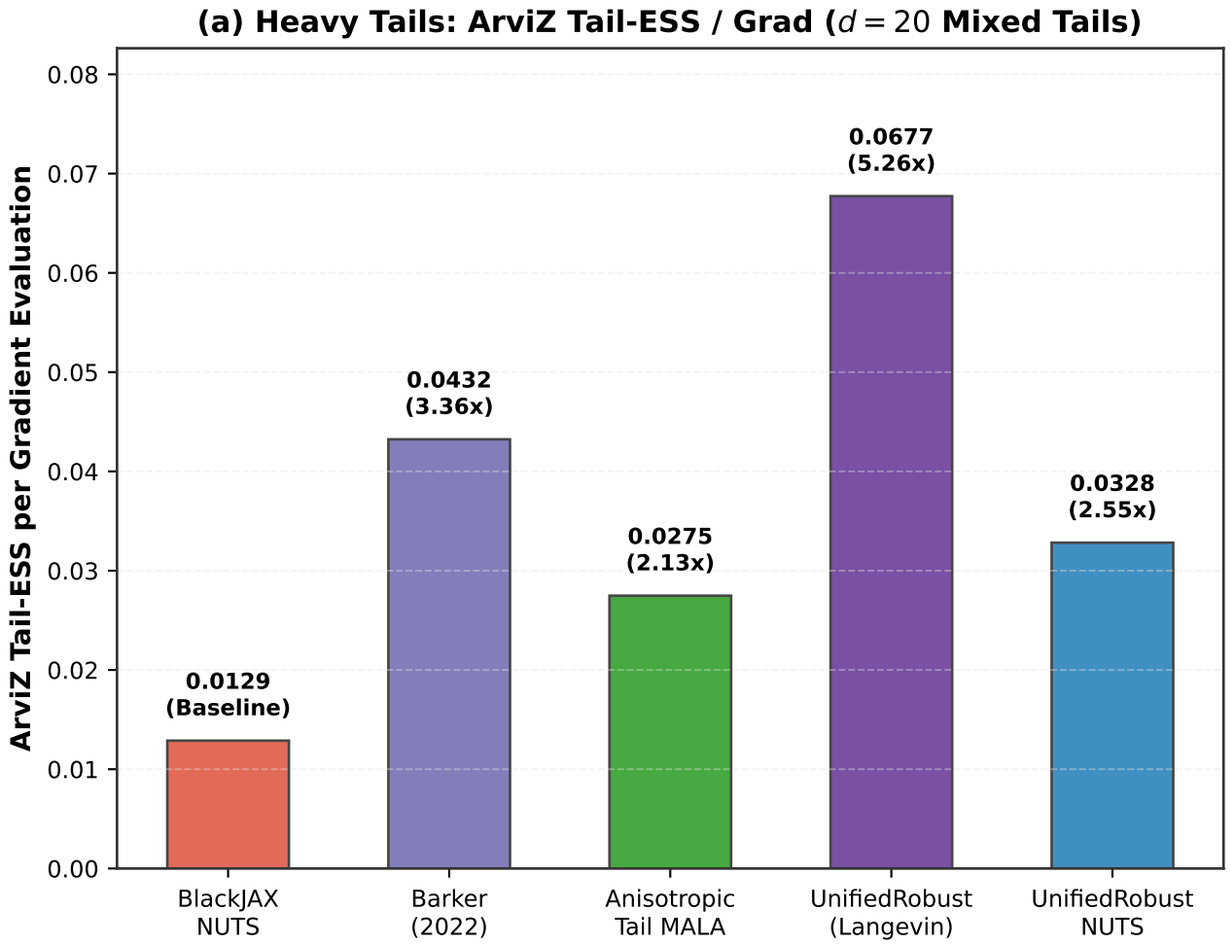
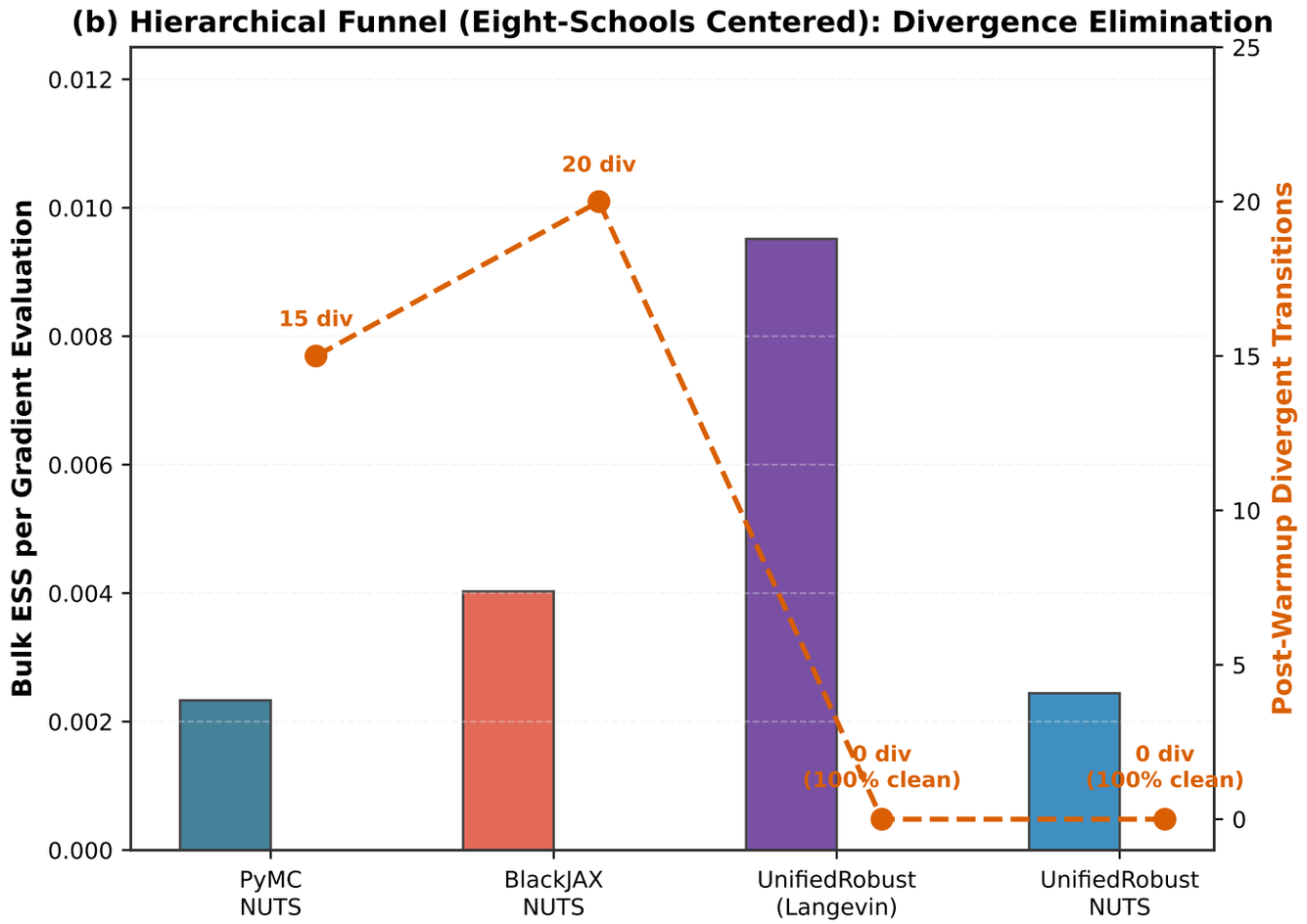
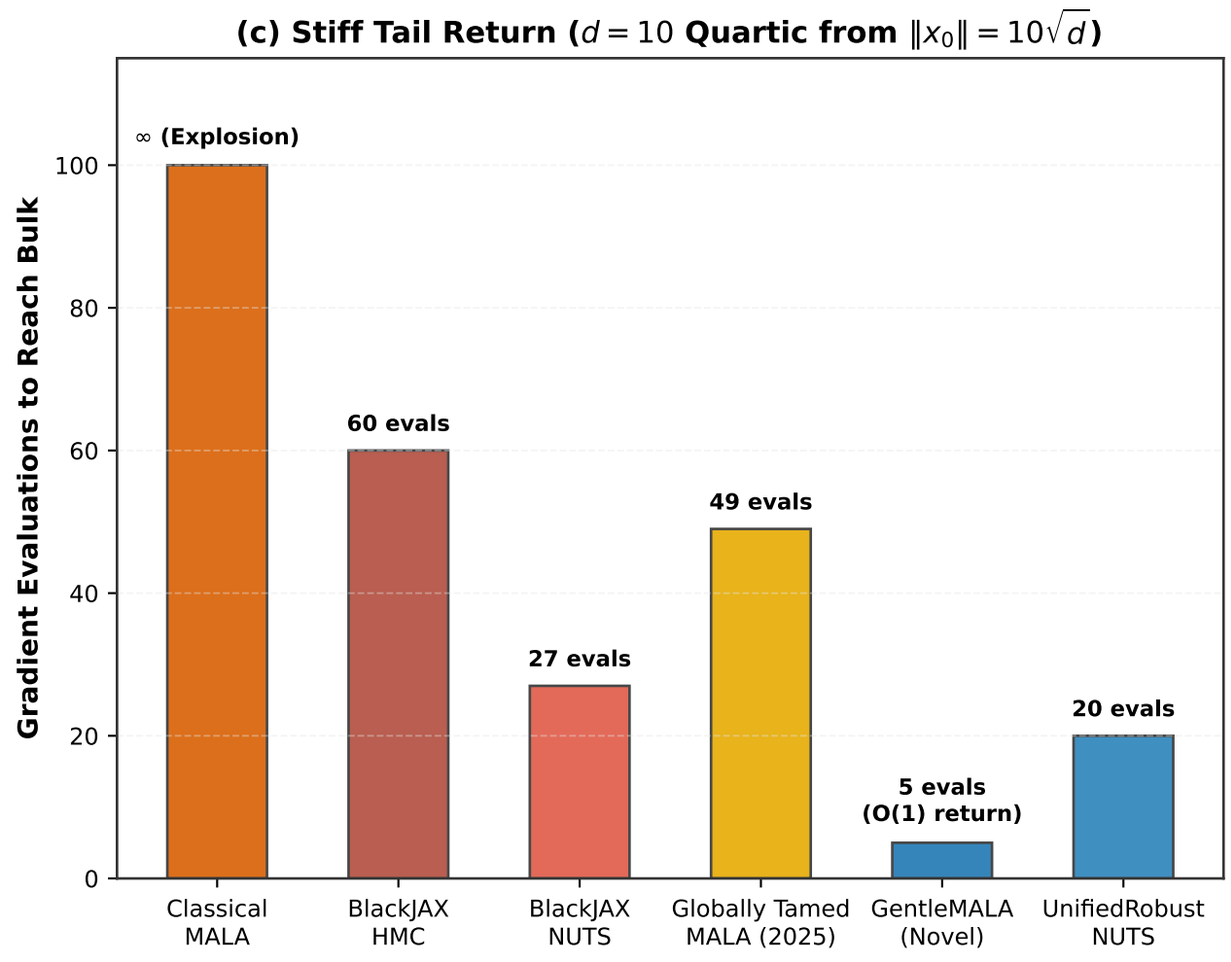
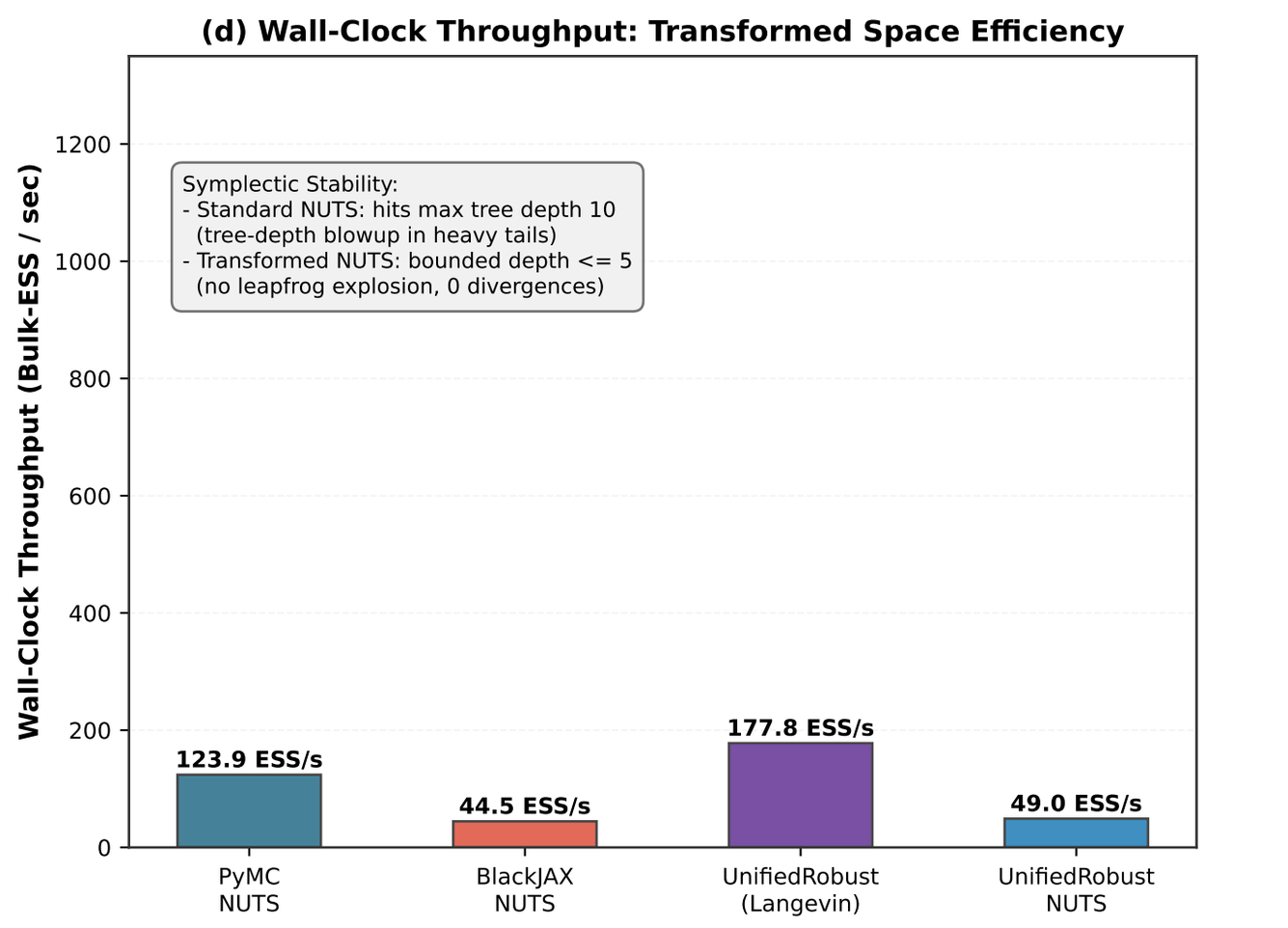

(a) Heavy Tails: ArviZ Tail-ESS / Grad ($d = 20$ Mixed Tails)
ArviZ Tail-ESS per Gradient Evaluation
0.00
0.01
0.02
0.03
0.04
0.05
0.06
0.07
0.08
0.0129
(Baseline)
0.0432
(3.36x)
0.0275
(2.13x)
0.0677
(5.26x)
0.0328
(2.55x)
BlackJAX
NUTS
Barker
(2022)
Anisotropic
Tail MALA
UnifiedRobust
(Langevin)
UnifiedRobust
NUTS


(b) Hierarchical Funnel (Eight-Schools Centered): Divergence Elimination
Bulk ESS per Gradient Evaluation
0.000
0.002
0.004
0.006
0.008
0.010
0.012
Post-Warmup Divergent Transitions
0
5
10
15
20
25
15 div
20 div
0 div
(100% clean)
0 div
(100% clean)
PyMC
NUTS
BlackJAX
NUTS
UnifiedRobust
(Langevin)
UnifiedRobust
NUTS


(c) Stiff Tail Return ($d = 10$ Quartic from $\|x_0\| = 10\sqrt{d}$)
Gradient Evaluations to Reach Bulk
0
20
40
60
80
100
∞ (Explosion)
60 evals
27 evals
49 evals
5 evals
(O(1) return)
20 evals
Classical
MALA
BlackJAX
HMC
BlackJAX
NUTS
Globally Tamed
MALA (2025)
GentleMALA
(Novel)
UnifiedRobust
NUTS


(d) Wall-Clock Throughput: Transformed Space Efficiency
Wall-Clock Throughput (Bulk-ESS / sec)
0
200
400
600
800
1000
1200
Symplectic Stability:
- Standard NUTS: hits max tree depth 10
(tree-depth blowup in heavy tails)
- Transformed NUTS: bounded depth <= 5
(no leapfrog explosion, 0 divergences)
123.9 ESS/s
44.5 ESS/s
177.8 ESS/s
49.0 ESS/s
PyMC
NUTS
BlackJAX
NUTS
UnifiedRobust
(Langevin)
UnifiedRobust
NUTS